Who Will Charge?

Modeling EV Charging Probability for KEBA AI Coding Contest Level 1

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The Problem

- The task was to predict whether a user is actively charging an EV at a specific hourly timestamp.
- For every row in the test set, we had to output a probability:

$$y_\text{prob} = P(\text{user is charging at this hour})$$

- The target variable was binary:

$$y_b = \begin{cases} 1, & \text{charging} \\ 0, & \text{not charging} \end{cases}$$

- The leaderboard metric was the **Brier Score**, so calibrated probabilities mattered more than hard 0/1 predictions.

Why Brier Score Matters

- Brier Score measures the mean squared error between predicted probabilities and true labels:

$$BS = \frac{1}{N} \sum_{i=1}^{N} (p_i - y_i)^2$$

- Lower is better.
- A prediction of 0.9 is excellent if the user charges, but very costly if the user does not.
- This pushed us to focus on **probability calibration**, not only classification accuracy.

Dataset Overview

- The training data contained hourly charging records for multiple users.
- Main columns:
  - `unique_id`: anonymized user identifier
  - `ds`: hourly timestamp
  - `y_b`: binary charging target
  - `y_e`: energy-related value, zero when not charging
- The test file contained future timestamps and required a `y_prob` prediction.
- The data was time-series data, so random train-validation splits would leak future behavior.

First Observation: Strong Class Imbalance

- Most hourly rows were non-charging rows.
- Charging happened only during a small fraction of all timestamps.
- This means a naive model can look reasonable by predicting low probabilities everywhere.
- But to score well, the model must still identify the few high-probability charging windows.

Key lesson: We needed a model that was conservative overall, but confident at the right hours.

Second Observation: Workplace Charging Pattern

- Charging was concentrated on weekdays.
- Weekends had very low charging activity.
- The strongest charging hours were usually morning to afternoon.
- This suggested a workplace or commuter-style charging pattern:
  - users arrive in the morning,
  - charging starts early,
  - charging continues for several hours.

Third Observation: User Behavior Was Not Equal

- Some users charged frequently.
- Some users almost never charged.
- A global average was not enough.
- The model needed to learn user-specific charging habits.

Example idea:

$$P(\text{charging} \mid \text{user}, \text{weekday}, \text{hour})$$

Fourth Observation: Calendar Effects

- Public holidays and bridge days had very low or zero charging.
- This was important because our first strong model still predicted normal weekday activity on some holidays.
- Manually correcting these dates improved the leaderboard score.
- Important corrected dates included:
  - Good Friday
  - Easter Monday
  - Labour Day
  - Ascension Day
  - likely bridge days after holidays

Feature Engineering

We created features in four main groups:

1. **Time features**
2. **User behavior features**
3. **Recent history features**
4. **Calendar correction features**

Time Features

- Hour of day: `hour`
- Day of week: `day_of_week`
- Month
- Weekend flag
- Business-hour flag

We also used cyclic transformations for time variables:

$$\sin\left(\frac{2\pi \cdot hour}{24}\right), \quad \cos\left(\frac{2\pi \cdot hour}{24}\right)$$

This helps the model understand that hour 23 and hour 0 are close to each other.

User Behavior Features

- User-level charging rate:

$$P(\text{charging} \mid \text{user})$$

- User-hour charging probability:

$$P(\text{charging} \mid \text{user}, \text{hour})$$

- User-weekday-hour probability:

$$P(\text{charging} \mid \text{user}, \text{weekday}, \text{hour})$$

These features captured the fact that every user had a different routine.

Recent History Features

- We used recent same-slot behavior:

  last k values for the same user, weekday, and hour

- Example:

  Was this user charging on the last few Tuesdays at 8 AM?

- We tested rolling windows such as 8, 12, 16, 20, and 26 weeks.
- Recent behavior was useful, but making the model too dependent on exact weekly repetition hurt the leaderboard score.

Calendar Features

- We added flags or manual corrections for:
  - weekends,
  - public holidays,
  - bridge days,
  - Easter-week reduction.
- These corrections were very important because the model otherwise treated holidays like normal working days.
- The biggest improvement came from reducing probabilities on holiday and Easter-related dates.

Modeling Approach

- We started with gradient boosting models such as XGBoost and LightGBM-style approaches.
- The main reason was that the data became tabular after feature engineering.
- Tree-based models are strong for:
  - nonlinear user behavior,
  - interaction between hour and weekday,
  - different charging habits per user,
  - sparse charging events.
- We then improved the submission using leaderboard-informed calibration and calendar corrections.

Validation Strategy

- We avoided random train-test splits because this is time-series data.
- Instead, we used chronological validation:
  - train on older data,
  - validate on later months.
- This better simulated the real task: predicting future charging behavior.
- We also used leaderboard feedback carefully to tune final probability calibration.

What Did Not Work

- Pure recent-week recurrence was too unstable.
- Scaling the wrong individual dates made the score worse.
- Increasing normal weekday probabilities also made the score worse.
- This showed that the model was already strong on normal days, while the main errors came from special calendar periods and calibration.

What Improved the Score

- Holiday and bridge-day masking improved the score.
- Easter-week probability reduction improved the score.
- Downscaling normal weekday probabilities slightly improved calibration.
- Leaderboard-informed blending gave our best result.

Main insight: The winning direction was not just a more complex model, but better probability calibration around special days.

Final Result

- Best Brier Score achieved:

  $\boxed{0.0257}$

- Level leaderboard position:

  $\boxed{22^{nd}}$

- Overall contest result:

  $\boxed{5^{th} \text{ overall}}$

This was a strong result because we improved the model through data understanding, calendar insight, and probability calibration.

Final Model Summary

Component	Purpose
Time features	capture daily and weekly cycles
User features	model individual charging habits
Same-slot history	capture recurring routines
Holiday corrections	reduce false positives on non-working days
Easter-week scaling	handle seasonal calendar effects
Probability calibration	optimize Brier Score

Key Takeaways

- Understanding the data was more important than blindly trying models.
- EV charging behavior was highly dependent on user, weekday, hour, and calendar context.
- Special days caused large prediction errors if treated as normal weekdays.
- Brier Score rewards calibrated probabilities, so small probability adjustments can matter.
- Our final improvement came from combining machine learning with domain reasoning.

Closing

We did not just predict charging.
We learned the rhythm of charging behavior.

Final Score: 0.0257
5th Overall